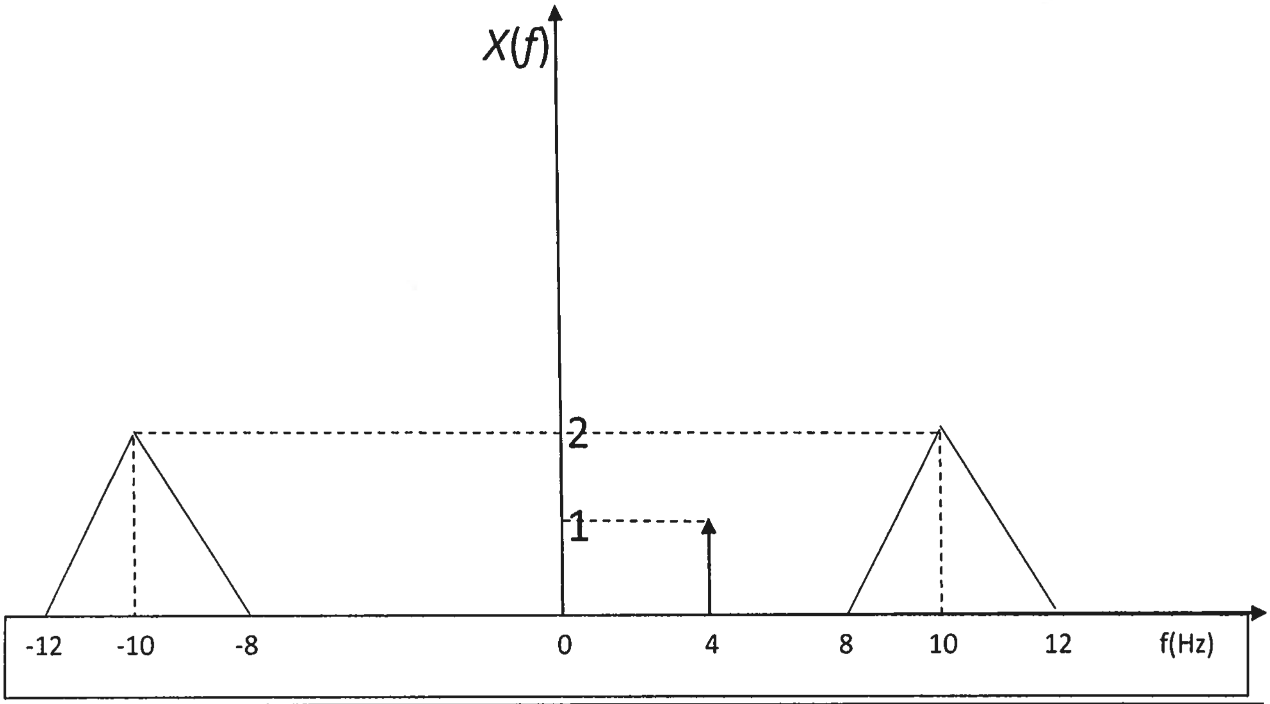
Quiz 1 Sample

Quiz1

Find the inverse Fourier transform of $X(f)$.



Note: (i) The figure is not drawn to scale.

(ii) The arrow at 4 Hz represents an impulse function.

Useful Information

$\mathrm{rect}(t) \rightleftharpoons \mathrm{sinc}(f)$

$\mathrm{triang}(t) \rightleftharpoons \mathrm{sinc}^2(f)$

$\mathrm{triang}(t) = 1 - |t| \text{ if } |t| < 1$

$\quad = 0 \text{ otherwise}$

If $g(t) \rightleftharpoons G(f)$ and $h(t) \rightleftharpoons H(f)$, then

$g(at) \rightleftharpoons \frac{1}{|a|} G(f/a)$

$G(t) \rightleftharpoons g(-f)$

$g(t - t_0) \rightleftharpoons G(f)\exp(-j2\pi f t_0)$

$g(t)\exp(j2\pi f_c t) \rightleftharpoons G(f - f_c)$

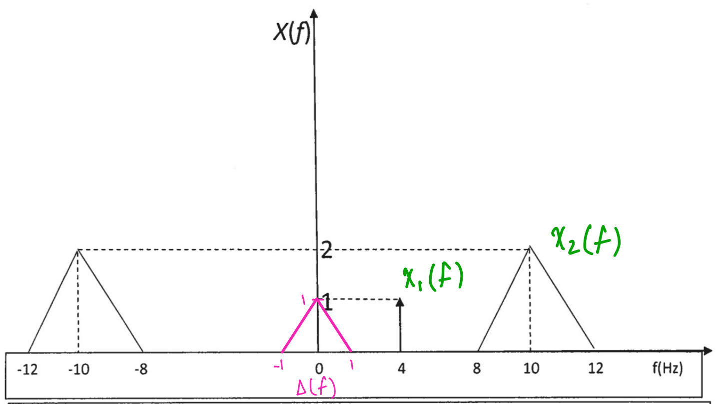
$g(t)\cos(2\pi f_c t) \rightleftharpoons \frac{1}{2}[G(f - f_c) + G(f + f_c)]$

$g(t) * h(t) \rightleftharpoons G(f)H(f)$

$g(t)h(t) \rightleftharpoons G(f) * H(f)$

Quiz 1 Sample

Find the inverse Fourier transform of $X(f)$.



Note: (i) The figure is not drawn to scale.

(ii) The arrow at 4 Hz represents an impulse function.

$t \rightharpoonup f$

$1 \longleftrightarrow \delta(f)$

$X_2(f) = 2\,\Delta(\tfrac{1}{2}f)$

$x_2(t) = 2^2 \operatorname{sinc}(2t)$

$= 4 \operatorname{sinc}(2t)$

$X(f) = X_1(f) + X_2(f)$

$X_1(f) = \delta(f-4) \longrightarrow x_1(t) = 1 \cdot e^{j2\pi 4t}$

$= e^{j8\pi t}$

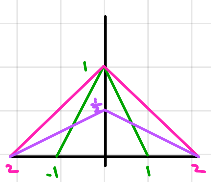
$X_2(f) = \frac{1}{2}\left[\Delta(f - \underset{\substack{\uparrow \\ f_c}}{10}) + \Delta(f+10)\right] \longrightarrow x_2(t) = 2\Delta^{-1}(f/2) \cdot \cos(2\pi f_c t)$

$= 2 \cdot 4 \operatorname{sinc}^2(2t) \cdot \cos(2\pi 10 t)$

$= 8 \operatorname{sinc}^2(2t) \cdot \cos(20\pi t)$

$X(f) = X_1(f) + X_2(f) \longrightarrow x(t) = x_1(t) + x_2(t)$

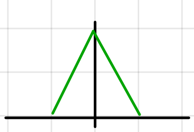
$\therefore x(t) = e^{j8\pi t} + 8 \operatorname{sinc}^2(2t) \cdot \cos(20\pi t)$



$\Delta(t) \rightharpoonup \operatorname{sinc}^2(f)$

$2\Delta(t) \rightharpoonup \overset{2}{\cancel{4}}\operatorname{sinc}^2(f)$?

$\frac{1}{2}\Delta(t) \rightharpoonup \overset{\frac{1}{2}}{\cancel{\tfrac{1}{4}}} \operatorname{sinc}^2(f)$?



scaling property: $x(at) \rightharpoonup \frac{1}{|a|} X\left(\frac{f}{a}\right)$

vs.

scalar multiplication: $k\,x(t) \rightharpoonup k\,X(f)$

$\therefore k\,x(at) \longleftrightarrow \frac{k}{|a|} X(f/a)$

$2\,x(2t) \longleftrightarrow \frac{2}{2} X(f/2)$ what

doesnt work like that

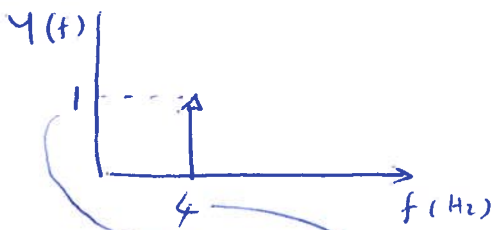
full width: τ rect(

QUIZ 1

SOLUTION:

SPLIT $X(f)$ INTO TWO PARTS:

$$X(f) = Y(f) + Z(f)$$



$$Y(f) = 1\cdot\delta(f-4)$$

$$\delta(f) \rightleftharpoons 1$$

FREQ. SHIFTING PROPERTY

$$\delta(f-4) \rightleftharpoons 1\cdot\exp(j2\pi\cdot 4\,t)$$

∴ LET $y(t) \rightleftharpoons Y(f)$

∴ $y(t) = \exp(j2\pi 4t)$

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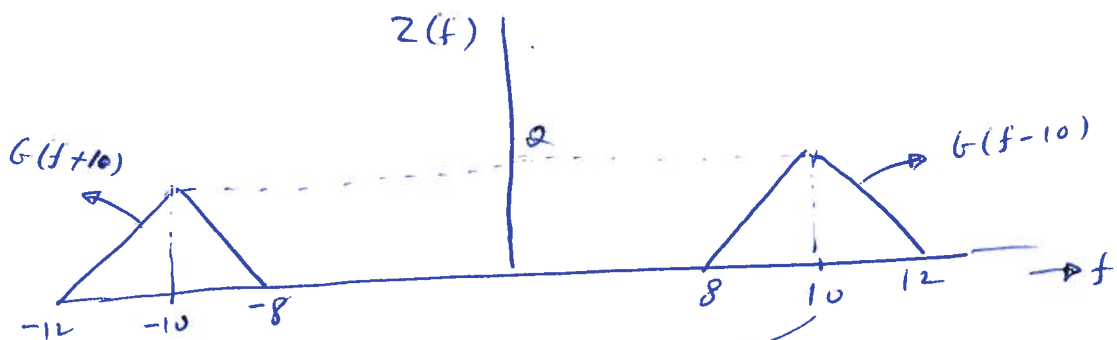
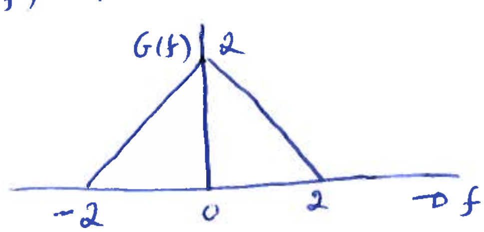
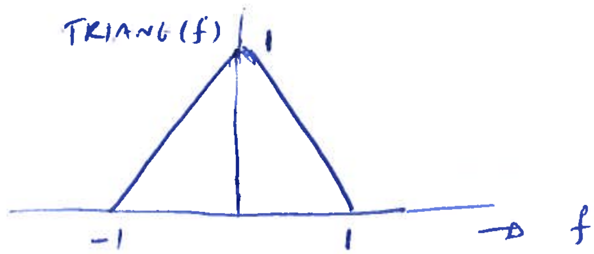


LET $Z(f) = G(f-10) + G(f+10)$

WHERE $G(f)$ IS THE BASEBAND SIGNAL SHOWN BELOW





WHEN $f = 1$, $\text{TRIANG}(f) = 0$

WHEN $f = 2$, $G(f) = 0$

$\therefore$ FREQ. SCALING FACTOR $= 2$

~~WHEN~~ AMP. OF $\text{TRIANG}(f) = 1$

AMP OF $G(f) = 2$

$\therefore$ AMP. SCALING FACTOR $= 2$

$$G(f) = 2\,\text{TRIANG}(f/2)$$

$$\text{TRIANG}(t) \rightleftharpoons \text{SINC}^2(f)$$

DUALITY PROPERTY $\text{SINC}^2(t) \rightleftharpoons \text{TRIANG}(-f) = \text{TRIANG}(f)$

SCALING PROPERTY $\text{SINC}^2(at) \rightleftharpoons \frac{1}{|a|}\,\text{TRIANG}(f/a)$

CHOOSE $a = 2$

$$\text{SINC}^2(2t) \rightleftharpoons \frac{1}{2}\,\text{TRIANG}(f/2) \longrightarrow (1)$$

WE NEED TO FIND THE INVERSE FOURIER TRANSFORM OF $G(f) = 2\,\text{TRIANG}(f/2)$. SO, MULTIPLY (1) BY 4 ON BOTHSIDES

$$g(t) = 4\,\text{SINC}^2(2t) \rightleftharpoons 2\,\text{TRIANG}(f/2) = G(f) \longrightarrow (2)$$

$$Z(f) = G(f-10) + G(f+10)$$

COROLLARY OF FREQ. SHIFTING PROPERTY:

$$g(t)\cos(2\pi f_c t) \rightleftharpoons \frac{1}{2}\left[G(f-f_c) + G(f+f_c)\right]$$

CHOOSE $f_c = 10$ Hz & MULTIPLY BY 2 ON BOTH SIDES,

$$z(t) = 2g(t)\cos(2\pi.10t) \rightleftharpoons G(f-10) + G(f+10) = Z(f)$$

FROM 2, WE HAVE $g(t) = 4\,\mathrm{sinc}^2(2t)$

$$\therefore z(t) = 2\cdot 4\cdot \mathrm{sinc}^2(2t)\cdot\cos(2\pi.10.t)$$

$$X(f) = Y(f) + Z(f)$$

$$\Rightarrow x(t) = y(t) + z(t)$$

$$= \exp(j2\pi.4t) + 8\,\mathrm{sinc}^2(2t)\cos(2\pi.10.t)$$

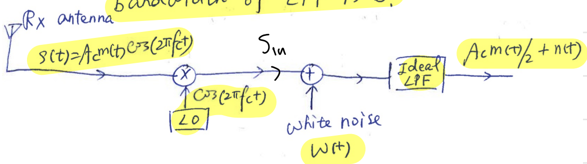
Quiz 2 Prep

$$SNR = \frac{P_{signal}}{P_{noise}} = \frac{S}{N}$$

$$SNR(dB) = 10\log_{10}\left(\frac{S}{N}\right) \rightarrow \text{power}$$
$$= 20\log_{10}\left(\frac{S}{N}\right) \rightarrow \text{voltage}$$

Steps
1) $N_{in} = 0 \rightarrow P_S = ?$
2) $S_{in} = 0 \rightarrow P_N = ?$
3) $SNR = \frac{P_S}{P_N}$

Q1) Problem 1: Calculate the SNR at the output of a DSB-SC receiver. Assume single-tone modulation $m(t) = A_m \cos(2\pi f_m t)$. Assume that the PSD of white noise is K, bandwidth of LPF is B.



- SNR = ?
- $N_{in} = 0 \rightarrow P_s = ?$

$$s(t) = A_c m(t)\cos(2\pi f_c t) \times LO$$
$$= A_c A_m \cos(2\pi f_m t)\cos(2\pi f_c t)\cdot\cos(2\pi f_c t)$$
$$= A_c A_m \cos(2\pi f_m t)\cos^2(2\pi f_c t) \leftarrow \cos^2(\theta) = \frac{1}{2}[1 + \cos(2\theta)]$$
$$= A_c A_m \cos(2\pi f_m t)\cdot\frac{1}{2}[1 + \cos(4\pi f_c t)]$$
$$= A_c A_m \cos(2\pi f_m t)\cdot\left[\frac{1}{2} + \frac{1}{2}\cos(4\pi f_c t)\right]$$
$$= \underbrace{\frac{1}{2}A_c A_m \cos(2\pi f_m t)}_{\text{baseband}} + \underbrace{\frac{1}{2}A_c A_m \cos(2\pi f_m t)\cos(4\pi f_c t)}_{\text{high freq.}}$$

$$P_s = \frac{1}{2}A^2$$
$$= \frac{1}{2}\left(\frac{1}{2}A_c A_m\right)^2$$
$$= \frac{1}{2}\cdot\frac{1}{4}A_c^2 A_m^2$$
$$= \frac{1}{8}A_c^2 A_m^2$$

- $S_{in} = 0 \rightarrow P_N = ?$

$$w(t) \rightarrow \boxed{LPF} \rightarrow N(t)$$

$$Y(f) = X(f)H(f)$$
$$S_N(f) = S_w(f)|H(f)|^2$$
$$= k\cdot \text{rect}\left(\frac{f}{2B}\right)$$

$$P_N = \int S_N(f)\,df$$
$$= \int_{-B}^{+B} k\cdot\cancel{\text{rect}\left(\frac{f}{2B}\right)}^{1}\,df$$
$$= k\cdot(B - -B)$$
$$= k2B$$
$$= 2kB$$

- $SNR = \frac{P_S}{P_N}$
$$= \frac{\frac{1}{8}A_c^2 A_m^2}{2kB}$$
$$\therefore = \frac{A_c^2 A_m^2}{16kB}$$

Q2) Problem 2

$X(t) = \underbrace{A_1 \cos(2\pi f_1 t)}_{X_1(t)} + \underbrace{A_2 \cos(2\pi f_2 t)}_{X_2(t)}$

$A_1 = 1V$, $A_2 = 0.5V$

$f_1 = 500Hz$, $f_2 = 1500Hz$

$w(t)$: white noise process with PSD $S_w(f) = 10^{-5}\, W/Hz$ $= k$

Bandwidth of the LPF $= 1kHz = B$

Find the SNR at the filter output.

$$SNR = \frac{P_S}{P_N} = ?$$

- $N_{in} = 0 \rightarrow P_S = ?$

$$x(t) = \overbrace{A_1 \cos(2\pi f_1 t)}^{x_1(t)} + \overbrace{A_2 \cos(2\pi f_2 t)}^{x_2(t)}$$

$$X(f) = \underbrace{\tfrac{1}{2}A_1[\delta(f-f_1) + \delta(f+f_1)]}_{X_1(f)} + \underbrace{\tfrac{1}{2}A_2[\delta(f-f_2) + \delta(f+f_2)]}_{X_2(f)}$$

$$\begin{aligned} Y(f) &= X(f)H(f) \\ &= [X_1(f) + X_2(f)]H(f) \\ &= X_1(f)H(f) + \cancel{X_2(f)H(f)} \\ &= X_1(f)H(f) \end{aligned}$$

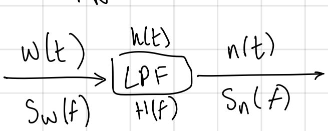
$\leftarrow$ $B_{LPF} = 1k$, $f_1 = 0.5k$, $f_2 = 1.5k$ Hz

$f_1 < \underbrace{B_{LPF} < f_2}_{f_2 \text{ cut out}}$

$$\begin{aligned} Y(f) &= X_1(f) \\ y(t) &= x_1(t) \\ &= A_1 \cos(2\pi f_1 t) \end{aligned}$$

$$\begin{aligned} P_S &= \tfrac{1}{2}A^2 \\ &= \tfrac{1}{2}(A_1)^2 \\ &= \tfrac{1}{2}A_1^2 \end{aligned}$$

- $S_{in} = 0 \rightarrow P_N = ?$



$$\begin{aligned} S_N(f) &= S_w(f)\, H(f) \\ &= k \cdot \text{rect}\left(\tfrac{f}{2B}\right) \end{aligned}$$

$$\begin{aligned} P_N &= \int S_N(f)\, df \\ &= \int_{-B}^{+B} k \cdot \cancel{\text{rect}\left(\tfrac{f}{2B}\right)}^{1}\, df \\ &= 2kB \end{aligned}$$

$A_1 = 1$ $\quad$ $k = \sim$

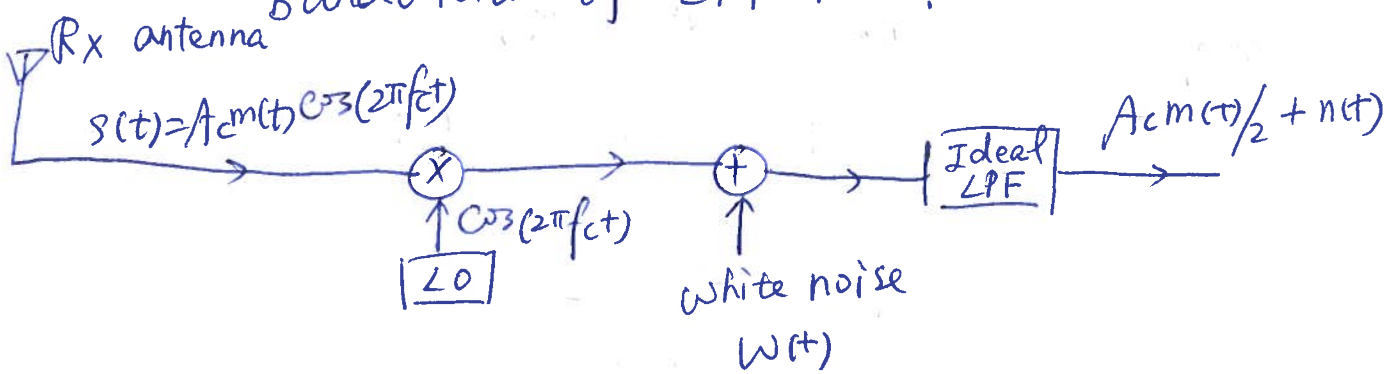
$$SNR = \frac{P_S}{P_N} = \frac{\frac{1}{2}A_1^2}{2kB} = \frac{1}{4kB} = X$$

$$SNR(dB) = 10\log_{10}(X) = y\ dB$$

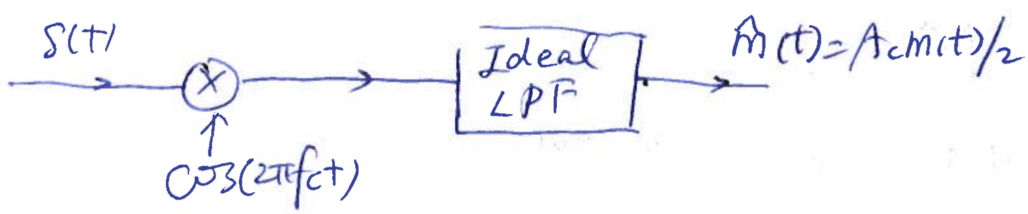
Problem 1: Calculate the SNR at the output of a DSB-SC receiver. Assume single-tone modulation

$m(t) = A_m \cos(2\pi f_m t)$.

Assume that the PSD of white noise is K, bandwidth of LPF is B.



Step 1: Turn off noise



$m(t) = A_m \cos(2\pi f_m t)$

Output of LPF $= \hat{m}(t) = \frac{A_c A_m}{2} \cos(2\pi f_m t)$

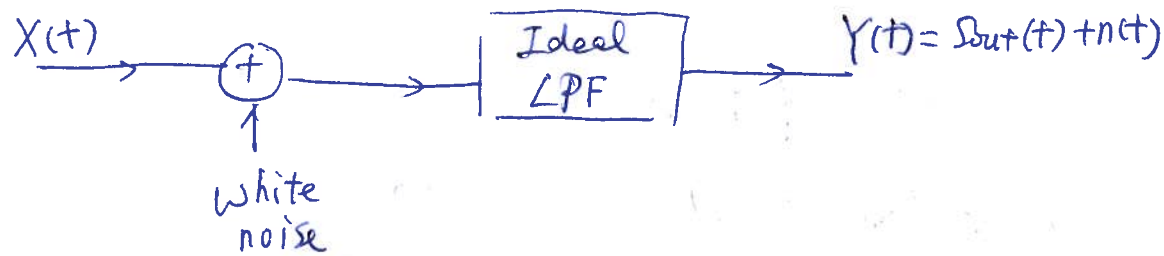
mean signal power $S = \frac{(A_c A_m/2)^2}{2} = \frac{A_c^2 A_m^2}{8}$ ①

Step 2: Turn off the signal

$S_N(f) = S_W(f)|H(f)|^2$

$S_W(f) = K$

Problem 2



$$X(t) = \underbrace{A_1 \cos(2\pi f_1 t)}_{X_1(t)} + \underbrace{A_2 \cos(2\pi f_2 t)}_{X_2(t)}$$

$A_1 = 1V$, $A_2 = 0.5V$

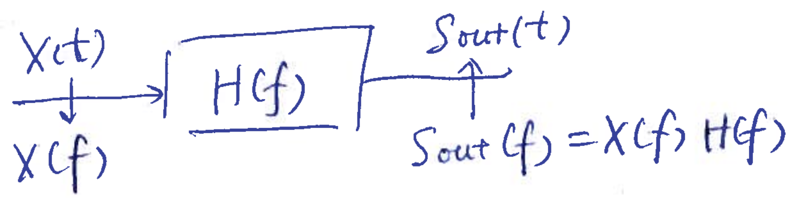
$f_1 = 500Hz$, $f_2 = 1500Hz$

$W(t)$: white noise process with PSD $S_W(f) = 10^{-5}\ W/Hz = k$

Bandwidth of the LPF $= 1kHz = B$

Find the SNR at the filter output.

Solution: Step 1 : signal flow (turn off noise)



$$X(t) = X_1(t) + X_2(t)$$

$$X_1(t) = A_1 \cos(2\pi f_1 t) \rightleftharpoons \frac{A_1}{2}[\delta(f-f_1) + \delta(f+f_2)]$$

$$= \frac{1}{2}[\delta(f-500) + \delta(f+500)]$$

$$= X_1(f)$$

$$X_2(t) = A_2 \cos(2\pi f_2 t) \rightleftharpoons \frac{A_2}{2}[\delta(f-f_2) + \delta(f+f_2)]$$

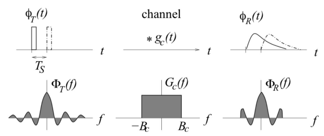
$$= \frac{0.5}{2}[\delta(f-1500) + \delta(f+1500)]$$

$$= X_2(f)$$

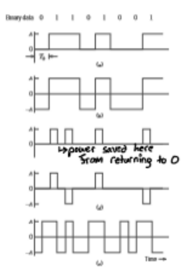
Quiz 3 Prep

Pulse Shaping

- Unless transmission symbol rate is very low, one cannot use impulse, narrow pulse or rectangular pulse to transmit data symbols
- Such pulses have very large (infinite) bandwidth, but the channel has finite bandwidth
- The pulse will spread out as their high-frequency components are suppressed, causing interference with neighbouring pulses (symbols)

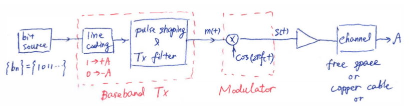


Line Codes (Cont.)



- **(a)**: unipolar NRZ signaling
- **(b)**: polar NRZ signaling
- **(c)**: unipolar RZ signaling
- **(d)**: bipolar RZ signaling
- **(e)**: slit-phase or Manchester code.

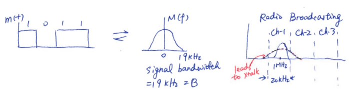
Digital Communication System with DSB-SC



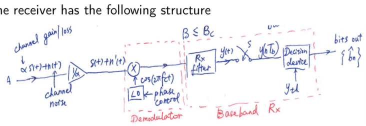
- The channel could be free space or copper cable or fiber optic cable
- The transmitter filter is used to limit the bandwidth of the transmitted signal so that it fits within the allocated channel bandwidth.
- In the example below, the allocated channel bandwidth B_c is less than the signal bandwidth. This leads to Xtalk

Digital Communication System with DSB-SC (Cont.)

- In the following , the allocated channel bandwidth B_c is less than the signal bandwidth. This leads to Xtalk

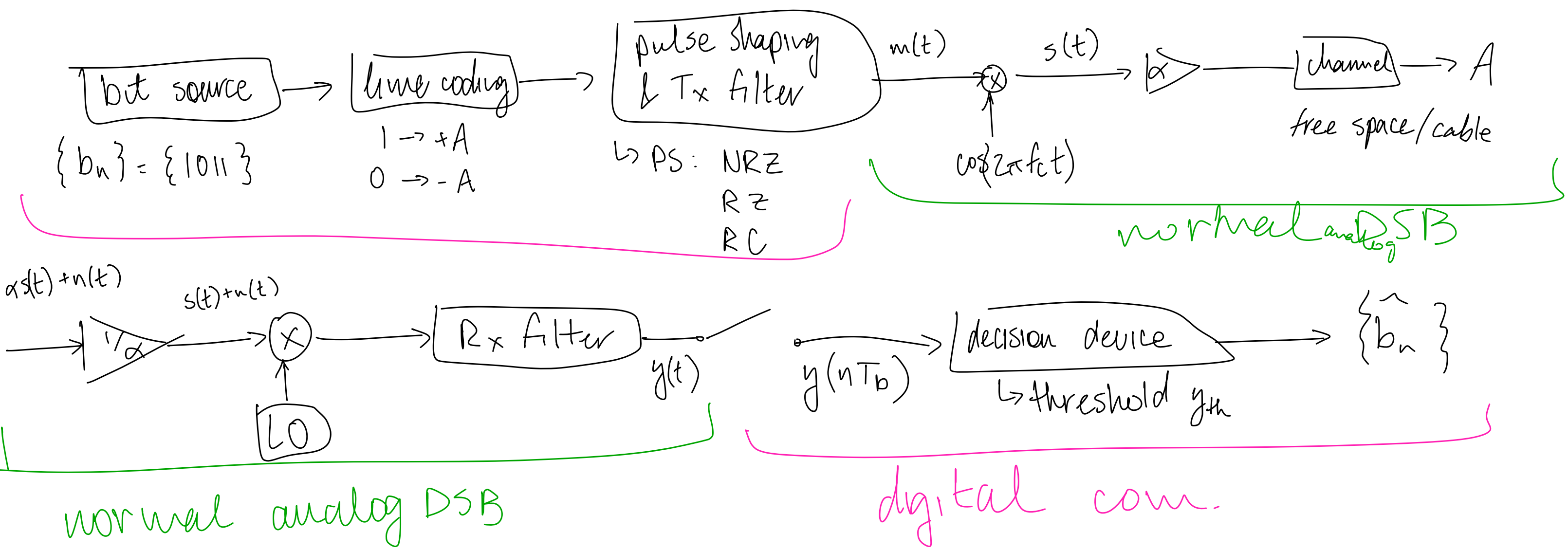


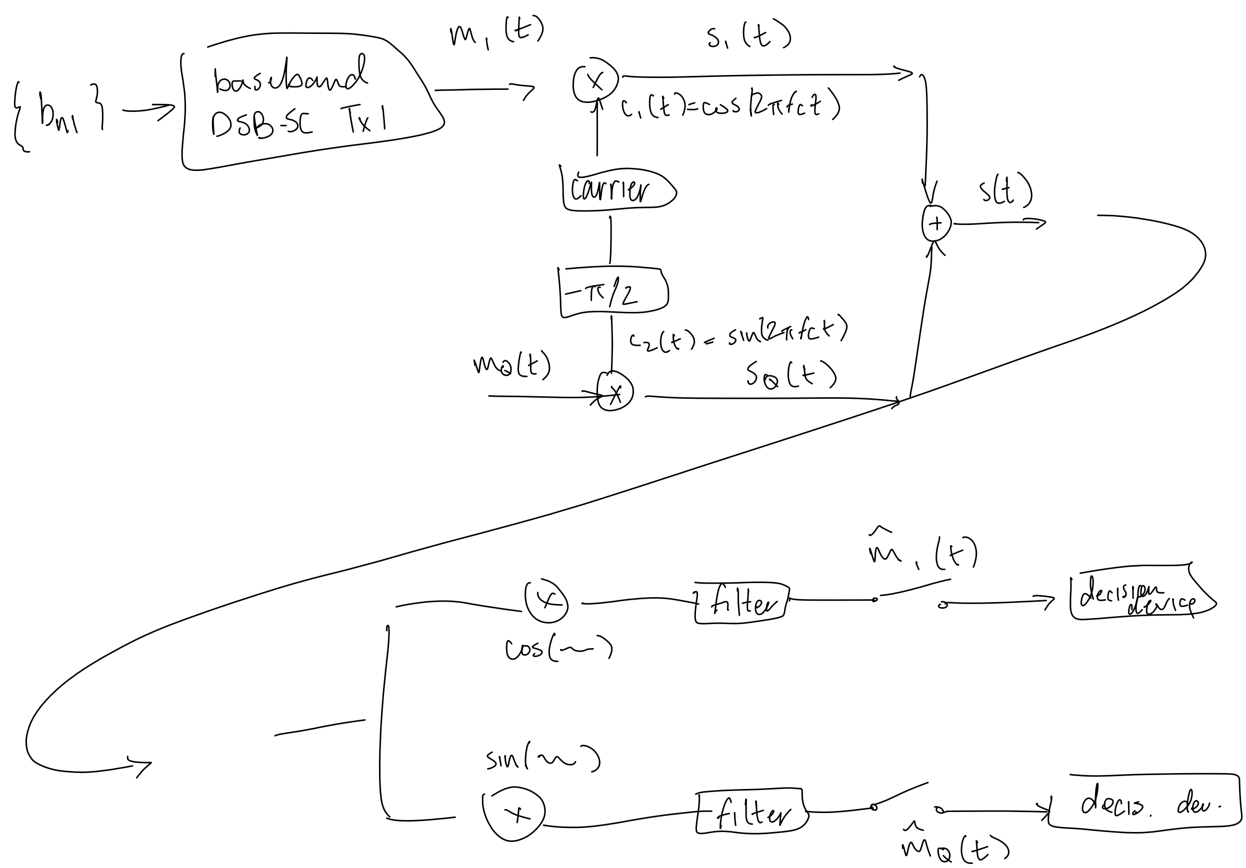
- The receiver has the following structure



Quiz 3

1. Draw a block diagram of a quadrature phase-shift keying (QPSK) digital communication system showing both the transmitter and receiver sections. Briefly explain the purpose of each block. Also explain the advantage(s) of QPSK over a BPSK modulation scheme. Draw the constellation diagrams of both BPSK and QPSK.





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	BPSK	QPSK
b/symbol	1 b/symbol	2 bit/symbol
BW efficency	k	x2 eff.
symbol rate		$\frac{1}{2}$ symbol rate of BPSK
power. eff.	slightly better for exact same	slightly worse but compensated by BW